CELLS, EMF, INTERNAL RESISTANCE

Chapter [3] - Current Electricity
Class - XII

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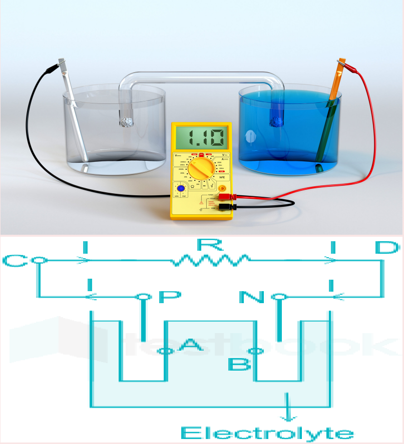
CELLS, EMF, INTERNAL RESISTANCE



Electric Cell

- **Electric Cell** is a **device** that can **generate electrical energy from chemical processes**.

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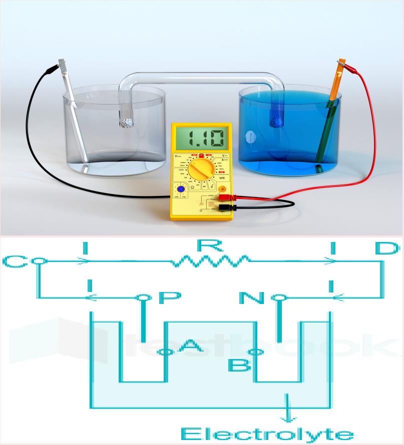

Electric Cell

- **Electric Cell** is a **device** that can **generate electrical energy from chemical processes**.
- Basically a **cell has two electrodes**, called the **positive (P)** and the **negative (N)**, They are immersed in an **electrolytic solution**.

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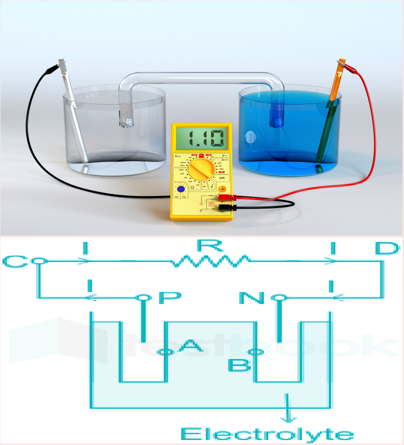

EMF

- When there is **no current (open circuit)**,the potential difference between the positive and negative electrode is called the **electromotive force (emf)** of the cell (ϵ).

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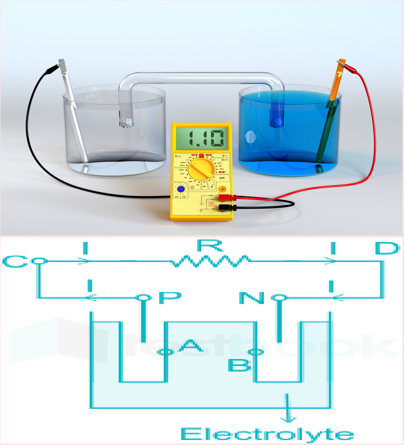

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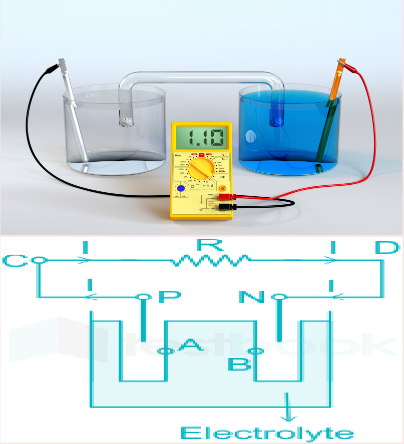

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- Note that **emf** is, actually, a **potential difference** and **not a force**.
- **Current flows** through the **electrolyte from N to P**, whereas through **R, it flows from P to N**.

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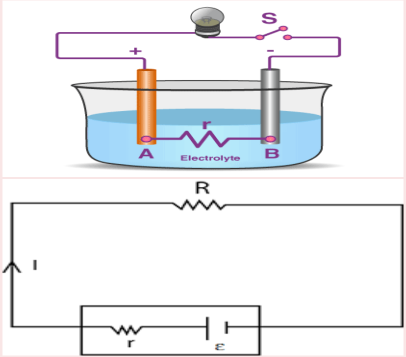

Internal Resistance Terminal Voltage

- The **electrolyte** through which a current flows **has a finite resistance r**, called the **internal resistance**.

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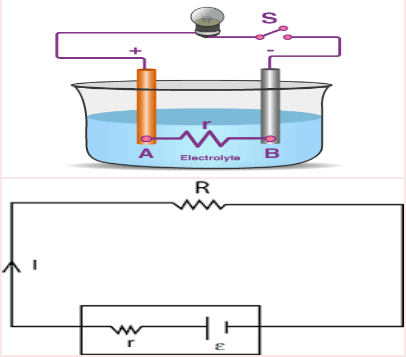

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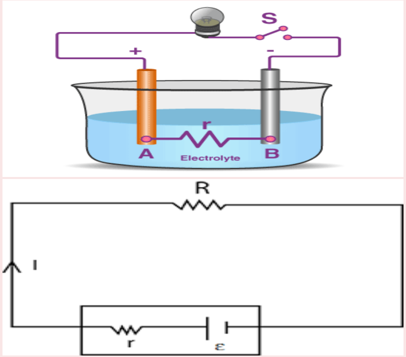

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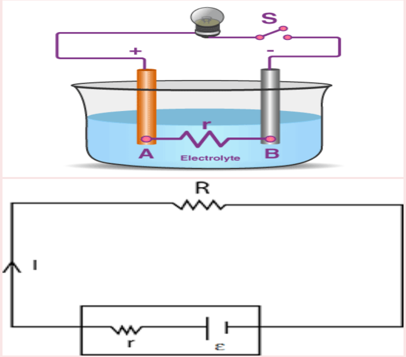

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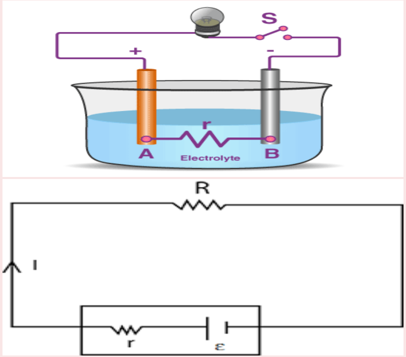

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$$\boxed{\epsilon - Ir = V} \qquad \because (V = IR)$$

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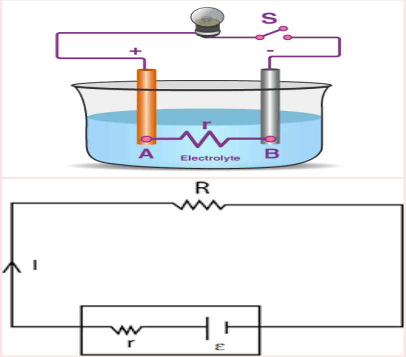

Internal Resistance and Terminal Voltage

- **If no current (I = 0)** is drawn from the cell (i.e., in an **open circuit**), the **potential difference V** across its terminal is **equal to its emf** ($\epsilon = V$)

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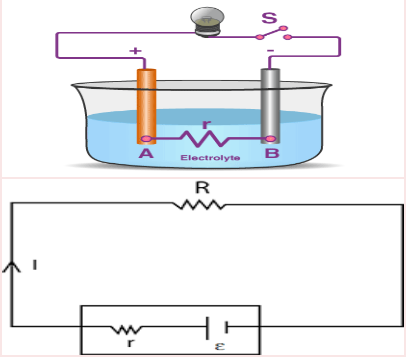

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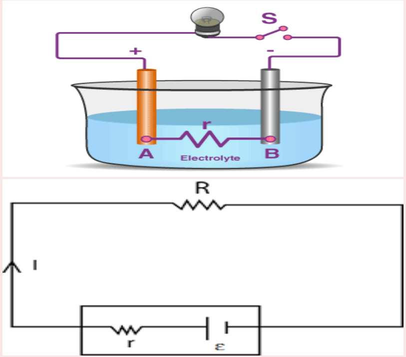

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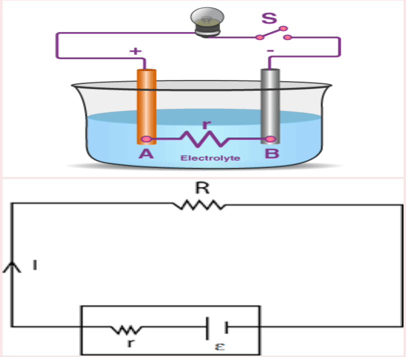

Internal Resistance and Terminal Voltage

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- if $I \neq 0$ V is less than ϵ
- From eq(1) We have $I = \dfrac{\epsilon}{R + r}$
- The **maximum current** that can be **drawn from a cell** is for $R = 0$ and it is $I_{max} = \dfrac{\epsilon}{r}$.

Numericals

Question 3.1

The storage battery of a car has an **emf** of 12 V. If the **internal resistance** of the battery is 0.4 Ω, what is the **maximum current** that can be drawn from the battery?

Numericals

Question 3.2

A battery of **emf 10 V** and **internal resistance** 3 Ω is connected to a resistor. If the **current** in the circuit is 0.5 A, what is the **resistance of the resistor**? What is the **terminal voltage** of the battery when the circuit is closed?

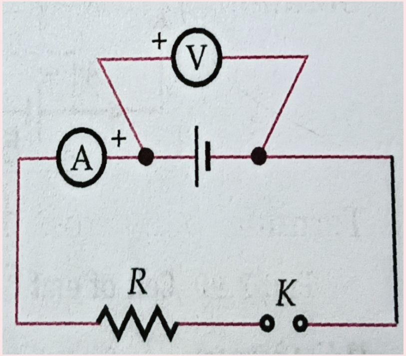
Numericals

Example 1

A voltmeter of resistance 998 Ω is connected across a cell of emf 2 V and internal resistance 2 Ω. Find the potential difference across the voltmeter, that across the terminals of the cell and percentage error in the reading of the voltmeter [CBSE OD 19]

Numericals

Example 2



In the circuit shown in fig., the voltmeter reads 1.5 V, when key is open. When the key is closed, the voltmeter reads 1.35 V and ammeter reads 1.5 A. Find the emf and the internal resistance of the cell.

Numericals

Example 3

A battery of emf 12 V and internal resistance 4 Ω is connected to an external resistance R. If the current in the resistance is 0.5 A, calculate the value of (a) R, and (b) the terminal voltage of the battery. [CBSE F 20]

Numericals

Example 4

A battery of e.m.f ξ and internal resistance 'r', gives a current of 0.5 A with an external resistor of 12 Ω and a current of 0.25 A with an external resistor of 25 Ω. Calculate (i) internal resistance of the cell and (ii) emf of the cell [CBSE D 02]

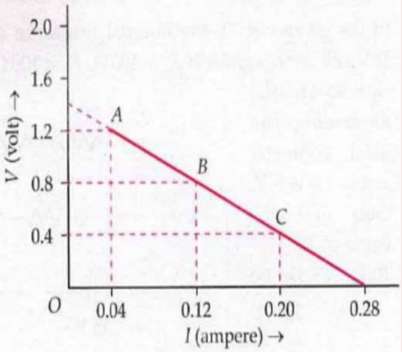
Numericals

Example 5

A battery of emf 3 volt and internal resistance r is connected in series with a resistor of 55 Ω throughan ammeter of resistance 1 Ω. The ammeter reads 50 mA. Draw the circuit diagram and calculate the value of r. [CBSE D 95]

Numericals

Example 6



Potential differences across terminals of a cell were measured (in volt) against different currents (in ampere) flowing through the cell. A graph was drawn which was a straight line ABC, as shown in Fig. Determine from the graph

(i) e.m.f. of the cell.

(ii) maximum current obtained from the cell, and

(iii) internal resistance of the cell [CBSE D 11C]

Numericals

Example 7

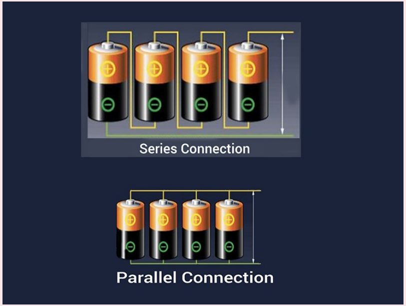
A cell of emf 6 and internal resistance r is connected across a variable load resistor R. It is found that when $R = 4\ \Omega$, the current is 1 A and when R is increased to $9\ \Omega$, the current reduces to 0.5 A. Find the values of the emf ξ and internal resistance r.

Numericals

Example 8

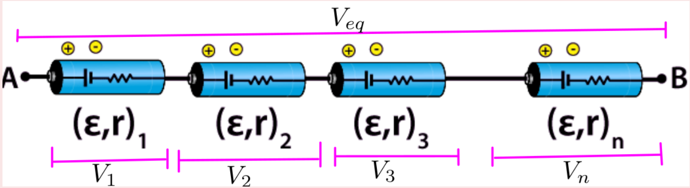
A cell of emf 4 V and internal resistance 1 Ω connected to a d.c. source of 10 V through a resistor of 5 Ω. Calculate the terminal voltage across the cell during charging.

CELLS IN SERIES AND IN PARALLEL



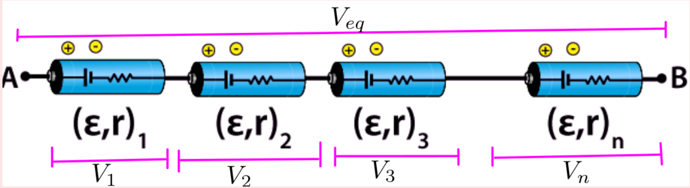
- A **cell is source of electric current**.
- A **single cell cannot give a strong current**. Therefore, **to get strong current two or more cells are to be combined**.
- The **combination of cells** is called a "**battery**".
- **Cells** can be **combined** in **three ways**:
  1. In **series**
  2. In **parallel** and,
  3. In mixed grouping.

CELLS IN SERIES



- Consider **n number of cells** are **connected in series**.
- Let $\epsilon_1,\ \epsilon_2, \cdots \epsilon_n$ are the **emf** of these cells and $r_1,\ r_2, \cdots r_n$ their **internal resistance** respectively.
- Let $(V_1 = \epsilon_1 - Ir_1),\ (V_2 = \epsilon_2 - Ir_2), \cdots (V_n = \epsilon_n - Ir_n)$ be the **potential difference** across each cells respectively.

CELLS IN SERIES



- Hence, the potential difference V_{eq} between the terminals A and B of the combination is

$$V_{eq} = V_1 + V_2 + \cdots + V_n$$
$$V_e q =$$